

LEDGER · 2026-08-28 · ADVANCED OLEOCHEMICALS

Enzymatic Synthesis of Estolide Biolubricants via Direct Castor Oil Transesterification

REJECTED

This concept did **not** clear the framework.

The reasoning is published in full below.

What the assessment found

Incumbent benchmark	Polyalphaolefin (PAO / PAO 4 / PAO 6) and Mineral Oil
Measured advantage	Wear scar diameter of 106 µm vs ~600 µm (standard diesel benchmark) / ~200 µm (PAO)
Time to first revenue	24 months
Regulatory risk	Moderate — qualification costs reported (\$750,000 estimated)

Why it failed

- Voided by: Mismatched comparison

Assessment

The concept fails sub-test B because the wear scar comparison (~600 µm baseline) relies on diesel fuel lubricity standards rather than a like-for-like fully formulated lubricant benchmark. Furthermore, mandatory OEM pump testing and REACH/VGP environmental certifications push time to first paid delivery beyond 18 months.

Also flagged

- Evidence rests on non-peer-reviewed sources
- Single-source evidence base
- no_production_runsheet
- no_equipment_list
- no_opex

A rejection is not a claim that the science is wrong. It means the venture case did not survive: the comparator was not what the buyer actually buys, the comparison was not like-for-like, the advantage fell short of a step change, or the capital and time to first revenue were prohibitive.

Assessed 2026-08-28 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

The Absence Audit

Method

Ledger

Free studies

Track record

Terms

Published for research and educational purposes. Nothing here is investment, legal, regulatory or engineering advice, and no commercial outcome is promised or implied. Concepts are drawn from published scientific literature and have not been commercially validated; several involve chemical, biological or regulated processes that require appropriate expertise, facilities, permits and safety controls. Independently verify every figure, citation and regulatory requirement before committing capital.

© 2026 The Absence Audit. All rights reserved. · [Terms of Use](#)